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Inventor(s)	Bloem; Michael et al.

Occupancy sensing systems and methods

Abstract

The present disclosure provides systems and methods for improved occupancy sensing. The methods and systems can deploy various signal threshold adjustments and/or signal analysis algorithms in response to sensed signals having a given quality, such as exceeding a threshold. In some cases, signal thresholds are lowered following an initial generated signal exceeding a first, higher threshold. In some cases, time-dependent signals are monitored using algorithms that analyze the signals for variations that are characteristic of human usage. Methods are disclosed for determining if two motion sensors are observing the same or overlapping spaces. Systems and methods for calibrating motion sensing systems are also disclosed.

Inventors: **Bloem; Michael (Grand Rapids, MI), Ward; Marcus (Portland, OR), Hall; Mychal (Portland, OR)**

Applicant: **Steelcase Inc.** (Grand Rapids, MI)

Family ID: **68532535**

Assignee: **Steelcase Inc. (Grand Rapids, MI)**

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Primary Examiner: Riddick; Blake C

Attorney, Agent or Firm: Quarles & Brady LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/474,495, filed on Sep. 14, 2021, which is a continuation of U.S. patent application Ser. No. 16/415,647, filed on May 17, 2019, now U.S. Pat. No. 11,125,907, issued Sep. 21, 2021, which claims priority to U.S. provisional application Ser. No. 62/673,776, filed on May 18, 2018, all of the contents of which are incorporated herein in their entirety by reference.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH
(1) N/A
BACKGROUND

(2) Occupancy and presence sensing technology is presently deployed in a variety of contexts. Most individuals are familiar with a passive infrared (PIR) sensor that controls room lights. Many of those individuals have experienced the situation where sitting still for too long causes the lights to shut off. The response is for the individual to wave their arms to provide some motion for the PIR sensor to trigger reactivation of the lights. Similarly, many of those individuals have experienced the situation where merely walking past a room is sufficient to trigger a PIR sensor into determining that a space is occupied, thus causing the lights to turn on despite the absence of any individuals within the space. In the context of lighting, this level of occupancy sensing is sufficient, because the cost (brief lack of lighting and having to wave arms or brief presence of lighting where it is not needed) is relatively inconsequential. However, with the advent of more advanced workspaces that include more complex automated processes, such as room reservations that can automatically adjust based on the occupancy status of a given location, activity/productivity tracking/monitoring that can make determinations regarding how people are utilizing various spaces and/or affordances, and other such processes, more accurate occupancy determinations are needed than those provided by traditional occupancy sensing systems.
(3) Accordingly, a need exists for occupancy sensing systems and methods that are robust in terms of their ability to accurately sense occupancy and are also low cost to make and use.

BRIEF SUMMARY

(4) The present disclosure provides systems and methods for improved occupancy sensing.
(5) In one aspect, the present disclosure provides a method of operating a motion sensor system. The motion sensor system includes a motion sensor that generates a motion signal in response to a sensed motion in a space to which the motion sensor is associated. The method includes: a) operating the motion sensor in a high threshold operating mode, wherein the high threshold operating mode includes a first signal threshold; and b) subsequent to step a), in response to the motion signal exceeding the first signal threshold, operating the motion sensor in a low threshold operating mode, wherein the low threshold operating mode includes a second signal threshold that is lower than the first signal threshold.
(6) In another aspect, the present disclosure provides a method of operating a motion sensor system. The motion sensor system includes motion sensor that generates a motion signal in response to a sensed motion in a space to which the motion sensor is associated. The method includes: a) operating the motion sensor in a high threshold operating mode, wherein the high threshold operating mode includes a first signal threshold; b) subsequent to step a), in response to the motion signal exceeding the first signal threshold, operating the motion sensor with a threshold adjustment function having a time-dependent signal threshold, wherein the time-dependent signal threshold is lower than the first signal threshold for at least a portion of the threshold adjustment function.
(7) In yet another aspect, the present disclosure provides a method of operating a motion sensor system. The motion sensor system includes a motion sensor that generates a motion signal in response to a sensed motion in a space to which the motion sensor is associated. The method includes: a) acquiring the motion signal from the motion sensor over a predetermined length of time, thereby providing a time-dependent motion signal; b) processing the time-dependent motion signal using a human-like motion analysis algorithm that distinguishes between motions having a time-varying component that is characteristic of human movement; and c) in response to the human-like motion analysis algorithm determining that a motion is characteristic of human movement, providing an occupancy signal indicative of occupancy of a space to which the motion sensor is associated.
(8) In a further aspect, the present disclosure provides a motion sensor system. The motion sensor system includes a motion sensor, a processor, and a memory. The memory has stored thereon instructions that, when executed by the processor, cause the processor to execute the methods described herein.

(9) In yet another additional aspect, the present disclosure provides a method of operating a motion sensor system. The motion sensor system includes a motion sensor having a fall sensor and at least one other motion sensor. The method includes: a) making a determination of motion within a space using the motion sensor system by receiving motion signals from the motion sensor; and b) in response to receiving a fall signal from the fall sensor that is representative of a falling motion, discontinuing utilizing the motion signals in making the determination of motion of step a), while continuing making the determination of motion utilizing other motion signals from the at least one other motion sensor.
(10) In yet a further aspect, the present disclosure provides a method of sensing motion using a motion sensing system including two or more motion sensor. The method includes: a) for each of the two or more motion sensors, generating a motion confidence index that includes a probability that a given motion sensor is sensing a desired motion of interest; b) sending the motion confidence index for each of the two or more motion sensors to a processor; and c) using the motion confidence index for each of the two or more motion sensors, determining whether or not the desired motion of interest occurred.
(11) In another aspect, the present disclosure provides a method of identifying relative positioning, location, and/or orientation of two or more motion sensors of a motion sensing system. The method includes: a) acquiring motion sense signals with the two or more motion sensors; b) comparing the acquired motion sense signals to determine if the two or more motion sensors are acquiring the same motions; and c) in response to the comparing of step b), deducing information regarding the relative positioning and/or orientation of the two or more sensors.
(12) In yet another aspect, the present disclosure provides a method of tuning an occupancy determination algorithm to a specific space having known characteristics. The method includes: a) receiving the known characteristics of the specific space; and b) adjusting one or more parameters of the occupancy determination algorithm in response to the known characteristics.
(13) In an additional aspect, the present disclosure provides a motion sensor calibration tool for calibrating a motion sensor system. The motion sensor system includes a motion sensor having a field of view, a processor, and a memory. The motion sensor calibration tool includes a sensor signal generator. The memory has stored thereon instructions that, when executed by the processor, cause the processor to initiate a signal generation algorithm in the sensor signal generator that, when the sensor signal generator is located the field of view of the motion sensor, produces a predictable motion signal from the motion sensor, and further calibrate the motion sensing system based on the predictable motion signal.
(14) In another aspect, the present disclosure provides a method of calibrating a motion sensing system including a motion sensor. The method includes: a) emitting an electromagnetic radiation of a known magnitude at a known location or moving a motion target at the known location, the known location positioned within a field of view of the motion sensor; b) measuring a calibration motion signal using the motion sensor during the emitting or moving of step a); and c) calibrating the motion sensing system using the calibration motion signal and the known location.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic representation of a system in accordance with the present disclosure.
(2) FIG. **2** is a space showing deployment of a system in accordance with the present disclosure.
(3) FIG. **3** is a space showing deployment of a system in accordance with the present disclosure.
(4) FIG. **4** is a space showing deployment of a system in accordance with the present disclosure.
(5) FIG. **5** is a space showing deployment of a system in accordance with the present disclosure.
(6) FIG. **6** is a space showing deployment of a system in accordance with the present disclosure.

(7) FIG. **7** is a top-down view of a space, as described in Example 1.
(8) FIG. **8** is a baseline noise signal, as described in Example 1.
(9) FIG. **9** is a view of sensor coverage of a space, as described in Example 1.
(10) FIG. **10** is a view of sensor coverage of a space, as described in Example 1.
(11) FIG. **11** is a plot of continuous movement monitoring, as described in Example 1.
(12) FIG. **12** is a plot of low movement signals from a human being, as described in Example 1.
(13) FIG. **13** is a plot of unoccupied baseline signals, as described in Example 1.

DETAILED DESCRIPTION

(14) Before the present invention is described in further detail, it is to be understood that the invention is not limited to the particular embodiments described. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting. The scope of the present invention will be limited only by the claims. As used herein, the singular forms “a”, “an”, and “the” include plural embodiments unless the context clearly dictates otherwise.

(15) It should be apparent to those skilled in the art that many additional modifications beside those already described are possible without departing from the inventive concepts. In interpreting this disclosure, all terms should be interpreted in the broadest possible manner consistent with the context. Variations of the term “comprising”, “including”, or “having” should be interpreted as referring to elements, components, or steps in a non-exclusive manner, so the referenced elements, components, or steps may be combined with other elements, components, or steps that are not expressly referenced. Embodiments referenced as “comprising”, “including”, or “having” certain elements are also contemplated as “consisting essentially of” and “consisting of” those elements, unless the context clearly dictates otherwise. It should be appreciated that aspects of the disclosure that are described with respect to a system are applicable to the methods, and vice versa, unless the context explicitly dictates otherwise.

(16) Numeric ranges disclosed herein are inclusive of their endpoints. For example, a numeric range of between 1 and 10 includes the values 1 and 10. When a series of numeric ranges are disclosed for a given value, the present disclosure expressly contemplates ranges including all combinations of the upper and lower bounds of those ranges. For example, a numeric range of between 1 and 10 or between 2 and 9 is intended to include the numeric ranges of between 1 and 9 and between 2 and 10.

(17) As used herein, “passive motion sensor” refers to a sensor that does not actively emit a signal that is used for sensing, but rather includes only a detection aspect for sensing a signal that is relevant to motion and/or presence. Examples of passive motion sensors include, but are not limited to, passive infrared sensors, capacitive proximity sensors, accelerometers, pressure sensors and other sensors known to those having ordinary skill in the art to function on similar principles to the passive infrared sensor.

(18) The present disclosure relates to systems and methods for monitoring motion, presence, occupancy, or the like using one or more passive motion sensors. The passive motion sensors can be space sensors, which are configured to monitor motion within a broader space. The passive motion sensors can be boundary sensors, which are configured to monitor motion within a narrower space to determine when a boundary is penetrated.

(19) The present disclosure also relates to systems and methods for monitoring motion, present, occupancy, or the like using one or more motion sensors that can be active or passive motion sensors. Again, these sensors can be space sensors, boundary sensors, or other sensors known to those having ordinary skill in the art.

(20) Referring to FIG. **1**, this disclosure provides a system **10** for monitoring occupancy of a space. The system can include one or more space sensors **12**, one or more boundary sensors **14**, and one or more gateways **16**. The system **10** can also include a processor **18** and a memory **20**. Each of the space sensors **12**, boundary sensors **14**, gateways **16**, processor **18**, and memory **20** can include a

wireless communication transceiver **22**. While a wireless communication embodiment is illustrated, any of the components of system **10** can be hardwired to one another as understood by those having ordinary skill in the art. In some cases, the system **10** can be deployed with one or more space sensors **12** and without any boundary sensors **14**. In some cases, the system **10** can be deployed with one or more boundary sensors **14** and without any space sensors **12**. In some cases, the system **10** can be deployed without the one or more gateways **16**. In those cases, the one or more space sensors **12** and the one or more boundary sensors **14** can be configured to communicate directly with the processor **18**. In some cases, the one or more space sensors **12**, the one or more boundary sensors **14**, and the one or more gateways **16** can include a processor **18** and a memory **20** locally co-housed with the respective sensors and/or gateways.

(21) Referring to FIG. **2**, one exemplary aspect of the system **10** is illustrated. In this aspect, the space **24** is the area of the room that is capable of being monitored by a single passive sensor. The system **10** include a single passive sensor in the form of a space sensor **12**. The space sensor **12** is mounted above the door **26**.

(22) Referring to FIG. **3**, one exemplary aspect of the system **10** is illustrated. In this aspect, the space **24** is the entire area of a room. The system **10** includes two space sensors **12** mounted in corners of the room (note: more or less space sensors **12** can be deployed and they can be placed in different locations, depending on the desired sensing properties). In this illustrated aspect, the only way into our out of the room is via a door **26**. A boundary sensor **14** is mounted in the vicinity of the door or within a part of the door, such as the door frame, and is configured to project a boundary field **28** into the doorway. The boundary sensor **14** is configured to sense when the boundary field **28** is penetrated by an object.

(23) Referring to FIG. **4**, another exemplary aspect of the system **10** is illustrated. In this aspect, the space **24** is a portion of the area of the room. The system **10** includes one space sensor **12** mounted directly above the space **24**. In this illustrated aspect, the space **24** can be entered from any horizontal direction (i.e., it cannot be entered from above, due to the ceiling, or below, due to the floor). A boundary sensor **14** is mounted and configured to project a boundary field **28** that covers at least a portion of the boundary of the space **24**. In the illustrated aspect, the boundary sensor **14** projects a cylindrical boundary field **28** that covers the entirety of the boundary of the space. In other aspects, the system **10** can include more than one boundary sensor **14** that each projects a boundary field **28** that covers a portion of the boundary of the space, with the total coverage provided by the multiple boundary fields **28** being sufficient to sense penetration of the boundary.

(24) Referring to FIG. **5**, another exemplary aspect of the system **10** is illustrated. In this aspect, the space **24** is an area within an open floor plan and the open floor plan includes multiple spaces **24**. The system **10** includes at least one space sensor **12** for each space **24**. The space sensor **12** can be configured and positioned to provide sufficient coverage of each space **24**. One or more boundary sensors **14** are mounted and configured to project one or more boundary fields **28** (not illustrated in this FIG. for ease of viewing, but would be projected along the vertical sides of each cube representing space **24**) that cover at least a portion of the boundary of the space **24**. In the illustrated aspect, the spaces **24** are square or block shaped and the boundary sensors **24** are configured to project the boundary fields **28** in a linear fashion along the edges of the square- or box-shaped spaces **24**. In this case, the system **10** can utilize a single boundary sensor **14** to sense the boundary of multiple spaces **24**, such as two neighboring spaces **24** that share a single boundary. The internal boundary sensors **13** are configured to monitor the boundary between two adjacent spaces **24** and the external boundary sensors **15** are configured to monitor the boundary between a space **24** and surrounding areas.

(25) Referring to FIG. **6**, another exemplary aspect of the system **10** is illustrated. In this aspect, the space **24** is a workstation bounded by partition panels **25**, and the specifically illustrated aspect shows a cubicle. The system can include at least one area of volume sensor **12** for the space **24**, such as the illustrated pressure-sensitive mat. One or more boundary sensors **14** can be mounted

and configured to project one or more boundary field **28** across an entrance to the cubicle. In some cases, the one or more boundary sensors **14** can be positioned to only monitor movement through the entrance to the cubicle, while in other cases, the one or more boundary sensors **14** can be positioned to monitor movement between the tops of the partition panels **25** and the ceiling.

(26) The space **24** can be a room, a workstation bounded by partition panels, such as personal semi-open working environment or a cubicle, an arbitrarily-defined area or volume, or another location where a determination of occupancy status is desired. It should be appreciated that, while the system **10** is described in the context of these spaces **24**, the inventive concept of the present disclosure can be used in any occupancy determination context. The concepts are applicable not only to spaces **24**, but any permanent object.

(27) The system **10** can include one or more space sensors **12** positioned in the space or locations near to the space. The one or more space sensors **12** can be configured to monitor an area or a volume within the space **24**. The one or more space sensors **12** can be oriented to monitor a portion of the space or the entire space. Examples of suitable space sensors **12** include, but are not limited to, passive infrared (PIR) sensors, floor pressure sensors, ultrasonic sensors, microwave sensors, other infrared sensors (e.g., thermopile, thermopile array, infrared camera, etc.), video sensors, acoustic sensors, and the like.

(28) The space sensors **12** can be battery operated. In some cases, the systems and methods described herein can extend battery lifetime by reducing the amount of communication signals necessary to make an occupancy determination.

(29) In cases where a single space sensor **12** is deployed, the space sensor **12** can be configured to provide high coverage of the space **24**, such as at least 90%, at least 95%, or at least 99% coverage of the space **24**, or low coverage of the space **24**, such as at most 50%, at most 25%, or at most 10% coverage of the space **24**.

(30) In cases where multiple space sensors **12** are deployed, the layout of the space sensors can be configured to provide high coverage of the space, such as at least 90%, at least 95%, or at least 99% coverage of the space, or low coverage of the space, such as at most 50%, at most 25%, or at most 10% coverage of the space **24**.

(31) It should be appreciated that the coverage described here is not only suitable for the function of the present disclosure, but in fact, the present disclosure contemplates low coverage situations where occupancy can still be accurately determined.

(32) It should be appreciated that some space sensors **12** can have “soft edges” to the area of coverage they provide. For example, 70% of a space can be covered extremely well, 20% can be covered somewhat well, and 10% can be covered poorly. For the purposes of defining coverage in the space sensor **12** context, a space sensor **12** can be considered to cover a volume if the space sensor **12** detects motion of a hand-sized object moving back and forth over a distance of 0.5 meters.

(33) In certain cases, the space sensor **12** can be a motion sensor that senses motion within the space.

(34) In certain cases, the space sensor **12** can be a pressure sensor, for example a floor or mat that is pressure sensitive, which senses pressure in a floor located in the space.

(35) In certain cases, the space sensor **12** can be an ultrasonic or Doppler sensor.

(36) The space **24** can have a boundary. In some cases, the boundary is defined by a door or other passageway, ingress, egress, or the like. In other cases, the boundary is an arbitrarily-defined area.

(37) The system can include one or more boundary sensors **14** positioned and configured to monitor penetration of at least a portion of the boundary. Examples of suitable boundary sensors **14** include, but are not limited to, curtain sensors, such as optical curtain sensors or magnetic curtain sensors, break-beam (also known as, “electric eye”) sensors, other infrared curtain sensors (e.g., thermopile, thermopile array, infrared camera, etc.), video sensors, and the like.

(38) In certain cases, the one or more boundary sensors **14** can be configured to provide at least

90%, at least 95%, or at least 99% coverage of the boundary of the space. For the purposes of defining coverage in the boundary sensor **14** context, a boundary sensor **14** can be considered to cover an area of the boundary if the boundary sensor **14** detects movement of a golf ball sized object through the boundary.

(39) The boundary sensors **14** can in some instances be directional boundary sensors that can sense the direction of movement through the boundary. For example, if two boundary sensors **14** are placed on the boundary with a very small distance separating the boundary sensors **14**, then the small difference in timing between sensing penetration can be used to determine the direction of travel.

(40) The space sensors **12** and boundary sensors **14** can be configured to communicate via a low energy consumption communication protocol, such as Bluetooth®, including Bluetooth® 4.0, Bluetooth® 4.1 Bluetooth® 4.2, Bluetooth® 5.0, Bluetooth® LE, and Bluetooth® Smart, IEEE802.15.4 (ZigBee®, 6LoWPAN, Thread®), via IEEE802.11 (WiFi), and the like. Of course, this protocol can be used with other, later issued low energy consumption communication protocols, as well as with other communication protocol specifications.

(41) The system can include one or more gateways **16** configured to send signals to and receive signals from the one or more space sensors **12** and the one or more boundary sensors **14**. In instances where the one or more gateways **16** include at least two gateways **16**, the at least two gateways **16** can be configured to communicate with one another via a low energy consumption communication protocols, such as Bluetooth®, via IEEE802.3 Ethernet, via IEEE802.11 (WiFi), and the like.

(42) In some cases, the sensors **12**, **14** can be integrated into a single unit. For example, a single unit residing above a door could monitor a space using a space sensor **12** and simultaneous monitor penetration of the door using a boundary sensor.

(43) In some cases, the sensors **12**, **14** and the communication capabilities of the gateway **16** can be integrated into a single unit.

(44) The system can include a processor **18** and a memory **20** that are in wired or wireless electronic communication with the gateway **16**. The processor **18** and memory **20** can be integrated into the gateway **16** or can be remote from the gateway **16**. In some cases, the processor **18** and memory **20** can be housed within the same facility as the sensors **12**, **14** and gateway **16**. In other cases, the processor and memory can be housed within a different facility than the sensors and gateway. For example, the processor **18** and memory **20** can be cloud computing systems.

(45) The processor **18** can be configured to communicate with the gateway via a low energy consumption communication protocol, such as Bluetooth®, via IEEE802.3 Ethernet, via IEEE802.11 (WiFi), and the like.

(46) The one or more space sensors **12**, one or more boundary sensors **14**, one or more gateways **16**, and processor **18** can be configured to execute a system communication protocol. For the simplicity of explanation, all space sensors **12** and boundary sensors **14** shall be referred to herein as “endpoints”. The system communication protocol will be described in some aspects using a singular endpoint and a singular gateway, but the concepts are applicable to multiple endpoints and gateways, unless the context clearly dictates otherwise.

(47) The motion sensors described herein can also include various fall sensors, orientation sensors, height sensors, and the like (not illustrated).

(48) The systems described herein can also include a motion sensor calibration tool. The motion sensor calibration tool can include a sensor signal generator. The sensor signal generator is an object that produces a known signal from a motion sensor when located at a known orientation relative to the motion sensor. For example, the sensor signal generator can be an electromagnetic radiation emitter that provides a known electromagnetic signal in a known direction, and which can provide a known response in a motion sensor given information regarding the relative positioning and orientation of the two. Referring to FIG. **2**, a sensor signal generator in the form of an

electromagnetic radiation emitter **30** is illustrated. As another example, the sensor signal generator can be an actuator (i.e., a linear actuator, rotational actuator, or the like) coupled to a motion target. The actuator can be activated to initiate a motion that is known. The known motion can provide a known response in a motion sensor given information regarding the relative position and orientation of the two. Referring to FIG. **3**, a sensor signal generator in the form of an actuator **32** coupled to a motion target **34** is illustrated.

(49) The motion sensor calibration tool also include a memory having stored thereon instructions that, when executed by the processor, cause the processor to initiate a signal generation algorithm in the sensor signal generator. The signal generation algorithm causes the signal sensor generator to initiate an emission or an action that produces a predictable motion signal from a motion sensor when the signal sensor generator is positioned within the motion sensor's field of view or in a known position within the motion sensor's field of view. The instructions can cause the processor to calibrate the motion sensing system based on the predictable motion signal. The instructions can further cause the processor to initiate acquisition of a baseline motion signal from the motion sensor in the absence of the signal generation algorithm and to calibrate the motion sensing system based on the baseline motion signal.

(50) In some cases, where the sensor signal generator is an infrared emitter, the sensor signal generator can emit a constant intensity infrared signal, a time-varying infrared signal, a pulsed infrared signal, or another infrared signal that has some characteristic property known to be useful for calibration in the sensing arts. The infrared emitter can be a point source that emits in substantially all directions or a directional source. The infrared emitter can be stationary or can be mounted to a moving arm, robot, or other mechanism by which a predictable movement can be initiated.

(51) Control of the sensor signal generator can be remote, for example, via a computing device, such as a laptop, smart phone, tablet, or the like, or can be local, for example, via a switch or a button on the sensor signal generator.

(52) The present disclosure includes various methods, as discussed below. These methods can be used in combination with various features of the systems described above or with other aspects of different methods described below, unless the context clearly dictates otherwise.

(53) In an aspect, the present disclosure provides a method of operating a passive motion sensor system. The passive motion sensor system includes a passive motion sensor. The passive motion sensor generates a passive motion signal in response to a sensed motion in a space to which the passive motion sensor is associate. The method includes: a) operating the passive motion sensor in a high threshold operating mode, wherein the high threshold operating mode includes a first signal threshold; and b) subsequent to step a), in response to the passive motion signal exceeding the first signal threshold, operating the passive motion sensor in a low threshold operating mode, wherein the low threshold operating mode includes a second signal threshold that is lower than the first signal threshold. In some cases, the method can include: a) operating the passive motion sensor in a high threshold operating mode, wherein the high threshold operating mode includes a first signal threshold; and b) subsequent to step a), in response to the passive motion signal exceeding the first signal threshold, operating the passive motion sensor with a threshold adjustment function having a time-dependent signal threshold, wherein the time-dependent signal threshold is lower than the first signal threshold for at least a portion of the threshold adjustment function. It should be appreciated that the low threshold operating mode is simply a threshold adjustment function that is a step function.

(54) When the passive motion signal exceeds the second signal threshold or the time-dependent signal threshold, the method can further include providing an occupancy signal indicative of occupancy of the space to which the passive motion sensor is associated. As processor that is co-housed with the passive motion sensor can execute the providing of the occupancy signal. The first signal threshold, the second signal threshold, and/or the time-dependent signal threshold can be

stored on a memory that is co-housed with the passive motion sensor.
(55) The second signal threshold can be less than 75%, less than 50%, less than 40%, less than 30%, less than 25%, less than 20%, less than 15%, less than 10%, less than 5%, or less than 1% of the first signal threshold. The time-dependent signal threshold can be less than 75%, less than 50%, less than 40%, less than 30%, less than 25%, less than 20%, less than 15%, less than 10%, less than 5%, or less than 1% of the first signal threshold for at least a portion of the threshold adjustment function.
(56) The threshold adjustment function can have a constant time-dependent signal threshold or can have a time-varying time-dependent signal threshold. The threshold adjustment function can be a step function having one, two, three, four, five, or more, up to N different time-dependent signal thresholds. In cases having multiple time-dependent signal thresholds, the time-dependent signal thresholds can increase over time.
(57) Once the system is operating in the low threshold operating more or with the threshold adjustment function, the methods can return to operating the passive motion sensor in the high threshold operating mode based on a variety of different conditions. For example, the methods can include returning to the high threshold operating mode after a given length of time has passed following the passive motion signal exceeding the first signal threshold. As another example, the methods can include returning to the high threshold operating mode after a given length of time has passed following the passive motion signal exceeding the second signal threshold or the time-dependent signal threshold. As yet another example, the methods can include both of these conditions for returning to the high threshold operating mode and can return based on whichever condition occurs later.
(58) In an aspect, the present disclosure provides a method of operating a passive motion sensor system. The passive motion sensor system includes a passive motion sensor. The passive motion sensor generates a passive motion signal in response to a sensed motion in a space to which the passive motion sensor is associate. The method includes: a) acquiring the passive motion signal from the passive motion sensor over a predetermined length of time, thereby providing a time-dependent passive motion signal; b) processing the time-dependent passive motion signal using a human-like motion analysis algorithm that distinguishes between motions having a time-varying component that is characteristic of human movement; c) in response to the human-like motion analysis algorithm determining that a motion is characteristic of human movement, providing an occupancy signal indicative of occupancy of a space to which the passive motion sensor is associated.
(59) The human-like motion analysis algorithm can include one or more of a frequency-analysis algorithm, a narrow bandpass filter configured to isolate frequencies that are characteristic of human movement, a transform of the time-dependent passive motion signal into a frequency-domain signal, a derivative of the time-dependent passive motion signal, and an integral of the time-dependent passive motion signal. The frequencies that are characteristic of human movement can be between 0.1 Hz and 2 Hz.
(60) In some cases, the methods can include of an environmental motion filter. The environmental motion filter can be essentially the opposite of the human-like motion analysis algorithm, where the time-varying component is characteristic of non-human movements.
(61) In some cases, the various signal thresholds and threshold adjustment functions can be stored locally in a memory that is co-housed with the passive motion sensor. In some cases, the various comparisons of signals with the various signal thresholds and threshold adjustment functions can be executed by a processor that is co-housed with the passive motion sensor. These two features afford energy efficient operation of the system. While memory retrieval and processing can consume energy, the relative energy cost of these functions is low compared with the energy cost of transmitting information, even when using a low energy transmission protocol, such as those described above.

(62) In certain aspects, the present disclosure provides a method of tuning a passive motion sensor or a passive motion sensor system. The method can include removing all motion-inducing elements from the room, providing a remote signal indicating that the room is unoccupied, recording a baseline signal, and then adjusting subsequent signals using the baseline signal.
(63) The method of tuning the passive motion sensor or the passive motion sensor system can also include intentionally inducing a signal or a motion for the purposes of tuning sensitivity of the sensor or system. The method can include use of the motion sensor calibration tool described above. The method can include: a) emitting an electromagnetic radiation (such as infrared radiation in the case of a PIR sensor) of a known magnitude at a known location or moving a motion target at the known location, the known location positioned within a field of view of the motion sensor; b) measuring a calibration motion signal using the motion sensor during the emitting or moving of step a); and c) calibrating the motion sensing system using the calibration motion signal and the known location. The method can also include: d) measuring a baseline motion signal using the motion sensor in the absence of the emitting or moving of step a); and c) calibrating the motion sensing system using the baseline motion signal. Steps d) and c) of this method can be executed as described above with respect to other aspects of methods of tuning the system or sensor.
(64) In an aspect, the present disclosure provides a method of sensing motion using a motion sensing system including two or more motion sensors. The method includes: a) for each of the two or more motion sensors, generating a motion confidence index that includes a probability that a given motion sensor is sensing a desired motion of interest (i.e., that it has not failed or lost accuracy) (using techniques such as resampling of time series to detect change points, or encoding sensor failure as a state in a dynamic Bayesian network or hidden Markov model representation of the sensing system, which is described in more detail in Example 6.5 of “Probabilistic Graphical Models: Principles and Techniques” by Koller and Friedman); b) sending the motion confidence index for each of the two or more motion sensors to a processor; and c) using the sensor readings and the motion confidence index for each of the two or more motion sensors, determining whether or not the desired motion of interest occurred. The motion confidence index can be computed by considering information such as a) data about sensor performance during a sensor calibration procedure, b) data about system performance provided by space users, c) data from other nearby sensors in the same space, d) data and models of “typical” sensor signals from “typical” motion, which could be collected during device development, and/or e) historical data from the sensor in question. This information can be tracked over time and any observed trends can be utilized in the motion confidence index.
(65) In an aspect, the present disclosure provides a method of identifying relative positioning, location, and/or orientation of two or more motion sensors of a motion sensing system. The method includes: a) acquiring motion sense signals with the two or more motion sensors; b) comparing the acquired motion sense signals to determine if the two or more motion sensors are acquiring the same motions (using techniques such as Pearson's or Spearman's cross-correlation coefficients for time series, time series clustering leveraging distance metrics such as Dynamic Time Warping, principal components analysis for time series, or clustering based on distances between the frequency domain representations of the time series signals from sensors, which can be computed using techniques like the fast Fourier transform); and c) in response to the comparing of step b) and any available information regarding the sensed space, inferring information regarding the relative positioning and/or orientation of the two or more sensors. In some cases, the acquire motion sense signals of step a) are acquired in response to initiating a known motion or a known emission of electromagnetic radiation. For example, the calibration methods described elsewhere herein with respect to the motion sensor calibration tool can be utilized in this method.
(66) The methods described above can in turn be utilized in broader methods of determining the probability that a space is occupied, such as rule-based algorithms with thresholds trained on historical ground truth data, physics-based approaches, supervised learning techniques that do not

explicitly consider the time series nature of the data (see, for example, “Elements of Statistical Learning” by Hastie et al.), probabilistic graphical models that explicitly consider the time series nature of the data (e.g., dynamic Bayesian networks or hidden Markov models, see, for example, “Probabilistic Graphical Models: Principles and Techniques” by Koller and Friedman), or neural network models that explicitly consider the time series nature of the data, such as the long short term memory recurrent neural network framework (see, for example, Hochreiter and Schmidhuber, “Long short-term memory,” Neural Computation 9.8 (1997), pages 1735-1780 and corresponding discussion in lecture 11 of the Stanford CS20SI TensorFlow course) or the gated recurrent units recurrent neural network framework (see, for example, Chung, Junyoung et al., “Empirical evaluation of gated recurrent neural networks on sequence modeling,” airXiv preprint airXiv: 1412.3555 (2014) and corresponding discussion in lecture 11 of the Stanford CS20SI TensorFlow course). These methods could be trained based on sensor and ground truth occupancy data collected before the deployment of sensors, based on additional data collected during a sensor calibration procedure, based on sensor data collected after the sensors have been deployed, and/or based on feedback from users of the outputs of the sensing system.
(67) In methods of occupancy determination, the various signals and occupancy determinations discussed above can be paired with additional information, such as scheduling information regarding the space, information regarding the type of space, information regarding the size of a space, individual location monitoring information, and the like.
(68) The present disclosure provides a method of tuning an occupancy determination algorithm to a specific space having known characteristics. The method includes: receiving the known characteristics of the specific space; and adjusting one or more parameters of the occupancy determination algorithm in response to the known characteristics.
(69) In certain aspects, where scheduling information is known, the processor or gateway can transmit a signal to the passive motion sensor to instruct the sensor to enter into a “sensitive” mode (i.e., operating in the low threshold operating mode or activating the threshold adjustment function).
(70) In certain aspects, where room-type information is known, the processor can make an occupancy determination based on a different set of rules for different rooms or room types. For example, if the system is deployed in a small room that is used as a quick huddle room or a room for phone conversations, then the occupancy determination can have less strict standards for releasing a room and indicating that the room is unoccupied. For a contrary example, if the system is deployed in a boardroom that is used for critical meetings only, then the occupancy determination can have more strict standards for releasing a room and indicating that the room is unoccupied.
(71) In addition to using signals from the sensors described herein, the methods can utilize other inputs, such as meta data associated with a given user, space, etc., information from a user's personal planner and/or calendar, such as meeting schedules, general availability, etc., and the like.
(72) In certain cases, the methods described herein can approximate the number of people in a given space.
Example 1
(73) An exemplary room and sensor arrangement is illustrated in FIG. **7**. The exemplary room was 211″×185″. The ceiling was 144″ high. The door was located along one of the 211″ walls. The temperature for these experiments was 72.5° F. The sensor was positioned at 9 feet height (unless indicated otherwise, as described below in the height-adjustment tests) and angled at 45° below horizontal. The sensor was a 1548 model PIR sensor, available commercially from Excelitas Technologies, Wheeling, Illinois, USA. A bandpass filter, 4-second window time, two pulse trigger, and 0.5-second blind time were used for experiments, unless otherwise indicated.
(74) Factors in Table 1 were controlled in experiments. Table 2 shows the results of a min/max screening test.

(75) TABLE-US-00001 TABLE 1 INPUTS DESCRIPTION Test Values PIR Voltage threshold for detection 0-1657.5 uV Sensitivity (50 uV Increments) PIR Pulse Number of pulses with a sign 1, 2, 3 Pulses Count change required to trigger motion (3 options) detection interrupt PIR Window Time in which pulses (pulse 2, 4, 6 Seconds Size counter) must occur for a motion (3 options) detection interrupt to be triggered Installation Installation height of the sensor on 7, 8, 9 Feet Height the wall. (3 options) External Major motion movement outside Movement, Movement the room to measure No Movement (Major impact of external movement. Test Motion subject shall be walking per NEMA standards

(76) TABLE-US-00002 TABLE 2 Est. of 1st Changing Physics SME Realistic Ease of Order Factor Units High Low High Low High Low Changing Impact PIR uV 1657.5 0 1250 30 400 60 High High Sensitivity PIR Pulse Pulses 4 1 3 1 2 1 High High Count PIR Sec 16 4 12 4 8 4 High High Window Size Installation Feet 12 0 9 7 8 8 Medium Medium Height External HBM Move- No Move- No Move- No High Medium Movement ment Move- ment Move- ment Move- (Major ment ment ment Motion)

(77) Table 3 shows factors that can be controlled, but which were held constant in experiments.

(78) TABLE-US-00003 TABLE 3 INPUTS DESCRIPTION PIR Blind Time Time for which sensor will ignore motion after it performs motion detection interrupt (actually after host MCU takes direct link high to low) PIR Op Mode Whether or not the sensor provides interrupts to the MCU PIR Filter Source No Testing: Voltage source provided when a data read is performed by the host MCU. When using the “Wake Up” op mode, BPF or LPF must be selected PIR High-Pass Cutoff Freq No Testing: High frequency cutoff for band-pass filter. (1548 only) PIR Pulse Detection Mode Whether or not a sign change is required when the signal exceeds the threshold (1548 only) to be considered a pulse. Sensor Presentation Angle Current testing will be done normal to the sensor. This assumes that the sensor lens is appropriately designed. It may be necessary to test the lens capabilities under various angles. Installation Height Current BKM installation is listed from 7-9 feet in increments of 1 foot. It is assumed that the installer has utilized the proper angle and tilt settings per design recommendations Room Size Four room sizes are determined for this project. Focus Room (<70 sq. ft [2 ppl]) Huddle Room (70-175 sq. ft [3-5 ppl]) Conference Room (175-350 sq. ft [6-10 ppl]) Ex. Lg Conference Room (350-560 sq. ft [5-16 ppl]) Unique Room Features Non-uniform variances in room types such as: Room dimensions/layout Furniture layout Double-wide Entrance Doors Unique window placement/number Manufacturers suggest mounting the PIRs in such a way that the PIR cannot “see” out of a window. Although the wavelength of infrared radiation to which the chips are sensitive does not penetrate glass very well, a strong infrared source (such as from a vehicle headlight or sunlight) can overload the sensor and cause a false alarm. A person moving on the other side of the glass would not be “seen” by the PID. That may be good for a window facing a public sidewalk, or bad for a window in an interior partition. HVAC Distance and Avoid mounting sensors close to air vents, as the vibration and air flow can Relative Temp reduce the effectiveness of the sensor (PIR sensors should not be within 4 ft of an air vent, and ultrasonic sensors should not be within 6 ft of an air vent). It is also recommended that the PIR not be placed in such a position that an HVAC vent would blow hot or cold air onto the surface of the plastic which covers the housing's window. Although air has very low emissivity (emits very small amounts of infrared energy), the air blowing on the plastic window cover could change the plastic's temperature enough to trigger a false alarm. Hot objects and moving air currents can affect the sensor's performance. Don't aim at air conditioners, heat vents, fireplaces, intermittent heat sources, etc. Sensor Type Corner, wall, curtain and ceiling brackets are currently utilized Room Temperature The sensor's performance depends on a differential between the ambient room temperature and the temperature of room occupants. Warmer rooms may reduce the sensor's ability to detect occupants. Direct Sunlight Care should be taken to avoid mounting in direct sunlight, however the degree to which direct sunlight affects the sensor is unknown at this time. Low Movement Non-movement or quick movements of the heat source inside the detection area. IR

Signature and Movement The IR signature of the person, distance, and actual movement are all factors Distance that enable the detection of movement by the PIR sensor

(79) Table 4 contains the values for the factors named in Table 3.

(80) TABLE-US-00004 TABLE 4 Constant Factor Units Constant Ease of changing Est. of 1st Order Impact PIR Blind Time Sec 2 seconds High Low PIR OP Mode NA Wake up Mode High Low PIR Filter Source NA Bandpass High Medium PIR High-Pass Cuttoff Freq Hz 0.4 Hz High Medium PIR Pulse Detection Mode NA +/− Required High None Sensor Presentation Angle Deg 45 deg Low Unknown Room Size NA Studio 1 used Low Unknown Unique Room Features NA Studio 1 used Low Unknown HVAC Distance Ft >4 feet High High HVAC Relative Temp Deg F. HVAC shall be verified off during testing and placed Low Unknown Low Movement NA Test Person: Mychal Hall Low Unknown Direct Sunlight NA Curtains shall be closed High Unknown Room Temperature Deg F. 73 deg F. Medium High Movement IR Distance Ft Test person shall be moving orthogonal to the sensor, Medium High at a test velocity of 1.0 m/s, at maximum distance Movement IR Signiture NA Test Person: Mychal Hall Medium High Sensor Type (Wall, Ceiling, Curtain) NA Wall Sensor High High

(81) Table 5 contains the results of screen test design, varying the factors described above in Table 1. The "pattern" column represents a value selection for the variables listed at the top of columns 2-6 of Table 5. A "+" sign indicates that the maximum value was used, a "−" sign indicates that the minimum value was used, and a "0" indicates that a middle value was used. Using the PIR Sensitivity as an example, the + corresponds to 1250, the − corresponds to 125, and the 0 corresponds to 687.5. Y-Major Motion Sensed and Notes columns represent outputs of the analysis. For each of the set of parameters investigates, no external movement was detected.

(82) TABLE-US-00005 TABLE 5 PIR Y-Major PIR PIR Pulse Window Install Ext. Motion Pattern Sensitivity Count Size Height Movement Sensed Notes ++−−− 1250 2 4 7 0 0 No External Movement Detected 0+00− 687.5 2 8 8 0 1 No External Movement Detected +−+−− 1250 1 12 7 0 1 No External Movement Detected 0−00− 687.5 1 4 9 10 0 No External Movement Detected +−−++ 1250 1 4 9 10 0 No External Movement Detected −−++− 125 1 12 9 0 1 No External Movement Detected −+−+− 125 2 4 9 0 1 No External Movement Detected −−−−+ 125 1 4 7 10 1 No External Movement Detected −++−+ 125 2 12 7 10 1 No External Movement Detected 0−00+ 687.5 1 8 8 10 1 No External Movement Detected +++++ 1250 2 12 9 10 0 No External Movement Detected

(83) Referring to FIG. **8**, a baseline noise spectrum was taken. The y-axis units are μV. When monitoring the baseline noise, it was noticed that the "waking up" of the PIR sensor contributes significantly to the noise levels. Increasing the blind time to 2 second significantly reduced this noise increase.

(84) A major motion test was performed with a 125 μV motion detection threshold and a 4 second blind. The major motion was measured in all locations in the room, save ˜2 square feet in the corners that touch the wall to which the sensor was mounted. A minor motion test was done in the seated locations, using the same parameters. Referring to FIG. **9**, the light gray represents areas of the room where major motion was detected and minor motion was not detected and the dark gray area represents areas of the room where both major and minor motion were detected.

(85) A second major and minor motion test was performed with a 60 μV motion detection threshold and a 4 second blind. The results are shown in FIG. **10**, with the same convention for light and dark gray areas. The minor motion is now detected in all seated positions, including those farthest from the sensor.

(86) Continuous data was collected for an occupied room over the course of 20 minutes. Varying thresholds were used to describe an event as an "occupied" event. Using a 200 μV threshold, 22 "occupied" events were registered in 20 minutes. Using a 400 μV threshold, 14 "occupied" events were registered in 20 minutes. Using a 800 μV threshold, the only events identified were when an occupant walked directly in front of the sensor when entering and leaving the room. These experiments illustrate feasibility of using a higher threshold to determine entrance and exit from the

space and a lower threshold to determine continued occupancy of the space, as described in various methods set forth above.

(87) A short time window of low-movement data for an occupied room is shown in FIG. **12** and a time window of the same length for an unoccupied room is shown in FIG. **13**. The data for the occupied room shows a time-dependent pattern that is characteristic of human movement. In addition, these data sets illustrated that a lower voltage threshold can be used to detect occupancy.

(88) While the above detailed description has shown, described, and pointed out novel features as applied to various embodiments, it will be understood that various omissions, substitutions, and changes in the form and details of the devices or algorithms illustrated can be made without departing from the spirit of the disclosure. As will be recognized, certain embodiments of the disclosures described herein can be embodied within a form that does not provide all of the features and benefits set forth herein, as some features can be used or practiced separately from others. The scope of certain disclosures disclosed herein is indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A method of calibrating a motion sensor system located within a facility, the motion sensor system including a motion sensor having a field of view (FOV), the method comprising the steps of: specifying a predictable motion signal that should be generated by the motion sensor upon detecting a first action that occurs within the motion sensor FOV; providing a sensor signal generator within the motion sensor FOV; causing the signal generator to complete the first action within the motion sensor FOV; obtaining a sensed signal via the motion sensor; and using a difference between the sensed signal and the predictable motion signal to calibrate the motion sensor.
2. The method of claim 1 wherein the step of causing the signal generator to complete the first action includes causing the signal generator to initiate a specific emission within the FOV.
3. The method of claim 1 wherein the step of causing the signal generator to complete the first action includes causing the generator to emit a known electromagnetic signal in a known direction.
4. The method of claim 1 wherein the step of causing the signal generator to complete the first action includes causing the generator to initiate a known motion within the FOV.
5. The method of claim 1 wherein the step of causing the signal generator to complete the first action includes causing the generator to initiate a known motion at a known location within the FOV.
6. The method of claim 1 wherein the step of causing the signal generator to complete the first action includes causing the generator to emit an infrared signal.
7. The method of claim 6 wherein the infrared signal is a time-varying infrared signal.
8. The method of claim 1 wherein the motion sensor is a passive motion sensor.
9. The method of claim 1 wherein the motion sensor is an active motion sensor.
10. The method of claim 1 further including, with no signal generator located within the motion sensor FOV, obtaining a baseline motion signal from the motion sensor and calibrating the sensing system using the baseline motion signal.
11. The method of claim 1 wherein the signal generator includes a mechanical assembly controlled to cause a known motion within the motion sensor FOV.
12. The method of claim 1 wherein the signal generator includes a robot that causes a known movement within the motion sensor FOV.
13. The method of claim 1 wherein the step of providing a sensor signal generator within the motion sensor FOV includes providing the sensor signal generator at a specific location within the motion sensor FOV.

14. A method of calibrating a motion sensor system located within a facility, the motion sensor system including a motion sensor having a field of view (FOV), the method comprising the steps of: specifying a predictable motion signal that should be generated by the motion sensor upon detecting a first action that occurs within the motion sensor FOV; providing a sensor signal generator within the motion sensor FOV; controlling the signal generator to cause at least one of a first motion within the motion sensor FOV and a first emission within the motion sensor FOV; obtaining a sensed signal via the motion sensor; and using a difference between the sensed signal and the predictable motion signal to calibrate the motion sensor.

15. The method of claim 14 wherein the step of providing a sensor signal generator within the motion sensor FOV includes providing the signal generator at a specific location within the FOV.

16. The method of claim 14 wherein the step controlling the signal generator to cause at least one of a first motion within the motion sensor FOV and a first emission within the motion sensor FOV includes controlling the signal generator to cause both of a first motion within the motion sensor FOV and a first emission within the motion sensor FOV.